6TH GRADE MATH · DATA DISPLAYS

Dot Plot Balance: Mean, Median & Spread

Explore & Reflect

Name _____ Date _____ Period _____

Open **Dot Plot Balance: Mean, Median & Spread** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Data: 5, 6, 6, 7, 8, 8, 9, 10, 13. If 13 changes to 20, which changes more, the mean or the median?

2. Where would you put a fulcrum (▲) under a dot plot so it balances?

2 Explore use the tool

1. Record the mean and median of the starting data.

2. Drag 13 to 20. Record the new mean and median.

3. Show the distances from the mean. Record the left and right totals for the starting data. Try another data set. What do you notice?

4. Make two data sets with mean 8: one with MAD less than 1 and one with MAD more than 3. Describe each dot plot.

3 Record use the tool and fill in the table

Build each data set (Remove values until 5 are left, then drag)

Quartiles: the median of each half, leaving out the middle value.

Data	Mean	Median	Range	IQR	MAD
4, 6, 8, 10, 12					
8, 8, 8, 8, 8					
2, 8, 9, 9, 12					
1, 2, 2, 3, 20					

4 Explain what did you discover?

1. The mean is the _____ point: the total distance below it equals the total distance _____ it.
2. An outlier pulls the _____ toward it more than the _____.
3. MAD is the average _____ from the mean.
4. Two classes both have a mean quiz score of 8. Class A has a MAD of 0.5 and Class B has a MAD of 3. What does that tell you about the two classes?

5 Apply a new situation, no tool

1. Find the mean and median of 3, 4, 4, 5, 34. Which one better describes a typical value? Why?
2. The mean of 4 numbers is 6. Three of them are 2, 5 and 9. Find the fourth.
3. Find the MAD of 2, 4, 6, 8.